Course Syllabus: Modern Data Warehousing & Engineering

Module 1: Foundations of Data Storage

Understanding why we moved from traditional systems to analytical powerhouses.

- **The Paradigm Shift:** Problems with traditional Database Systems and the shift to Data Warehousing.
- **Core Concepts:** Definition of a Data Warehouse (DWH)
- **Database vs. Data Warehouse:** Key architectural and use-case distinctions.
- **OLTP vs. OLAP:** Deep dive into the differences between transactional processing and analytical processing.

Module 2: Data Integration & Modern Architectures

How data moves and how we structure the journey from raw source to refined insight.

- **ETL & ELT Fundamentals:** Definitions, differences, and standard ETL layers.
- **Medallion Architecture:** Bronze, Silver, and Gold layer framework.
- **Core Concepts:** Understanding, USE CASES and Definitions
    - Data Marts
    - Data Warehousing
    - ACID Properties
- **Strategic Architectures:**
    - **Inmon:** The Top-Down approach (Enterprise DWH).
    - **Kimball:** The Bottom-Up approach (Data Marts).
- **The Modern Stack:**
    - Data Vault and key benefits
    - The rise of the **Data Lakehouse** and its characteristics
- **Comparison:** Lakehouse vs. Data Lake vs. Data Warehouse.

Module 3: Data Modeling Mastery

The "bread and butter" of data engineering—structuring data for performance and clarity.

- **Normalization:**
    - 1NF

        - 2NF
        - 3NF principles.

- **Modeling Types:** Relational vs. Dimensional vs. ER Models.
- **Phases of Modeling:** From Conceptual to Logical to Physical models.
- **Schema Design:** The Star Schema and Snowflake Schema.
- **The Building Blocks:** Facts vs. Dimensions.

Module 4: Advanced Dimensional Modeling

Handling complex business logic and changing data over time.

- **Dimension Deep Dive:**
    - Slowly Changing Dimensions (SCD Types 0, 1, 2, 3, 4).
        - SCD TYPE 0
        - SCD TYPE 1
        - SCD TYPE 2
        - SCD TYPE 3
        - SCD TYPE 4
    - Conformed Dimensions
    - Junk Dimensions
    - Shrunken Dimensions
    - Date Dimensions
    - Degenerate Dimensions
    - Role-Playing Dimensions.
- **Fact Table Specialized Types:**
    - Transactional
    - Periodic Snapshot
    - Accumulating Snapshot
    - Factless Facts.
- **Dimensional Data Modelling and Use Cases**
- **Integrity:** Referential Integrity and Constraints in a warehouse environment.

Module 5: Data Movement & Storage Formats

The technical execution of data pipelines and file-level optimization.

- **Loading Strategies:** Full Load vs. Incremental Load and the importance of **Idempotency**.
- **CDC (Change Data Capture):**

        - Time-based CDC
        - Trigger-based CDC
        - Log-based CDC.
- **File Formats:** Row-based (AVRO) vs. Columnar (Parquet).
- **The "Big Three" Table Formats:**
        - Delta Lake
        - Apache Iceberg
        - Apache Hudi

Module 6: Practical Application

Putting theory into practice with a real-world scenario.

- **Capstone Project:** Hands-on Hospital Dimensional Data Model.
- **Use Case Analysis:** Applying dimensional modeling to healthcare business processes.